TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	82.9 ± 4.39
Correlation Stability	92.6 ± 1.43
Distribution Consistency	87.0 ± 2.86
Entropy Preservation	76.8 ± 4.81
Mean Z Score Drift	0.1 ± 0.03
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	22.0 ± 3.71
Data Density	100.0 ± 0.00
Coverage Score	62.1 ± 3.62
Feature Accuracy	66.6 ± 9.55
Novelty Score	0.7 ± 0.02

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Dataset Audit: thoracic_surgery.csv

1. Dataset Portrait

Attribute	Value
Rows	470
Columns	17
Numeric features	17
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
DGN	3.096	3.106	+0.011	0.3%	OK
PRE4	3.282	3.230	-0.052	1.6%	OK
PRE5	4.569	10.252	+5.683	124.4%	Drift
PRE6	0.781	0.803	+0.022	2.9%	OK
PRE7	0.066	0.024	-0.042	63.7%	Drift
PRE8	0.145	0.136	-0.009	6.2%	OK
PRE9	0.066	0.098	+0.032	49.2%	Drift
PRE10	0.687	0.694	+0.007	1.0%	OK
PRE11	0.166	0.128	-0.038	23.1%	Warn
PRE14	11.736	11.662	-0.074	0.6%	OK
PRE17	0.074	0.066	-0.008	10.7%	Warn
PRE19	0.004	0.003	-0.002	37.5%	Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	82.9 ± 4.39
Correlation Stability (*2)	92.6 ± 1.43
Distribution Consistency (*3)	87.0 ± 2.86
Entropy Preservation (*4)	76.8 ± 4.81
Mean Z Score Drift (*5)	0.1 ± 0.03
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	22.0 ± 3.71

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	62.1 ± 3.62
Feature Accuracy (*10)	66.6 ± 9.55
Novelty Score (*11)	0.7 ± 0.02

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8436170212765959	0.8468085106382979	0.8489361702127666	0.005319148936170226	
F1	0.01111111111111111	0.021052631578947368	0.03111111111111111	0.02	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.21063829787234045	0.577659574468085	0.33404255319148934	0.12340425531914893	
F1	0.256855620661189	0.2559989983654612	0.260599923779373	0.00374430311818401 7	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8180851063829788	0.8351063829787234	0.8191489361702127	0.0010638297872340274	
F1	0.10369767110400277	0.12118250170881753	0.13233800079909992	0.028640329695096427	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8531914893617023	0.8542553191489363	0.8489361702127666	-0.004255319148936154	
F1	N/A	0.024444444444444442	0.01111111111111111	0.01111111111111111	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8553191489361703	0.8553191489361703	0.8553191489361703	0	
F1	N/A	0.0	N/A	0	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.

ACCURACY	0.851063829787234	0.8531914893617023	0.8521276595744682	0.0010638297872340385	
F1	0.037675637056442016	0.057817337461300314	0.060187847068652024	0.022512210012210012	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8138297872340425	0.8085106382978724	0.8159574468085108	0.002127659574468095	
F1	0.13319306364803737	0.13710389280183313	0.13149014994667169	-0.0017029137013657145	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8468085106382979	0.8521276595744682	0.8468085106382979	-1.1102230246251566e-17	
F1	N/A	0.04509803921568627	0.03398692810457517	0.03398692810457517	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8553191489361703	0.8553191489361703	0.8553191489361703	0	
F1	N/A	0.011111111111111111	N/A	0	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8393617021276596	0.8265957446808511	0.8351063829787234	-0.004255319148936154	
F1	0.07745532245532245	0.08701324389404884	0.15686788615581185	0.0794125637004894	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	2.0	2.88	2.16	1.0	0.0	0.0	0.0	1.0	1.0	14.0
2	3.0	3.4	1.88	0.0	0.0	0.0	0.0	0.0	0.0	12.0
3	3.0	2.76	2.08	1.0	0.0	0.0	0.0	1.0	0.0	11.0
4	3.0	3.68	3.04	0.0	0.0	0.0	0.0	0.0	0.0	11.0
5	3.0	2.44	0.96	2.0	0.0	1.0	0.0	1.0	1.0	11.0
6	3.0	2.48	1.88	1.0	0.0	0.0	0.0	1.0	0.0	11.0
7	3.0	4.36	3.28	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	2.0	3.19	2.5	1.0	0.0	0.0	0.0	1.0	0.0	11.0
9	3.0	3.16	2.64	2.0	0.0	0.0	0.0	1.0	1.0	11.0
10	3.0	2.32	2.16	1.0	0.0	0.0	0.0	1.0	0.0	11.0
11	3.0	2.56	2.32	0.0	0.0	1.0	0.0	1.0	0.0	12.0
12	3.0	4.28	4.44	1.0	0.0	0.0	0.0	0.0	0.0	12.0
13	3.0	3.0	2.36	1.0	0.0	0.0	0.0	1.0	1.0	11.0
14	2.0	3.98	3.06	2.0	0.0	0.0	0.0	1.0	1.0	14.0
15	3.0	1.96	1.4	1.0	0.0	0.0	0.0	1.0	0.0	11.0
16	3.0	4.68	4.16	1.0	0.0	0.0	0.0	1.0	0.0	12.0
17	2.0	2.21	1.88	0.0	0.0	1.0	0.0	0.0	0.0	12.0
18	2.0	2.96	1.67	0.0	0.0	0.0	0.0	0.0	0.0	12.0
19	3.0	2.6	1.68	1.0	0.0	0.0	0.0	1.0	0.0	12.0
20	3.0	2.88	2.48	0.0	0.0	0.0	0.0	0.0	0.0	11.0

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	5.0	1.44	0.96	2.0	0.0	0.0	0.0	1.0	0.0	12.0
2	3.0	3.704274991435156	2.8770264726725667	1.0	0.0	0.0	0.0	1.0	0.0	11.0
3	3.0	1.8996667485625125	1.2882419861996866	1.0	0.0	0.0	0.0	1.0	0.0	13.0
4	3.0	1.44	0.96	1.0	0.0	1.0	0.0	1.0	0.0	11.0
5	3.0	1.44	0.96	1.0	0.0	0.0	0.0	1.0	0.0	11.0
6	4.0	5.746573989161383	2.973625512562945	0.0	0.0	0.0	0.0	0.0	0.0	11.0
7	3.0	2.68	2.16	0.0	0.0	0.0	0.0	0.0	0.0	12.0
8	4.0	2.32	1.76	1.0	0.0	1.0	0.0	1.0	1.0	11.0
9	2.0	2.72	2.08	0.0	0.0	0.0	0.0	0.0	0.0	12.0
10	3.0	2.32	1.76	0.0	0.0	0.0	0.0	0.0	0.0	12.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	1.44	0.96	1.0	0.0	1.0	0.0	1.0	0.0	12.0
2	3.0	3.2655921508572496	2.490834702801637	1.0	0.0	0.0	0.0	1.0	0.0	11.0
3	3.0	2.173833826782977	1.4912364663764377	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	3.0	1.44	0.96	1.0	0.0	1.0	0.0	1.0	0.0	11.0
5	4.0	6.298472102068692	3.7751555135271033	0.0	0.0	0.0	0.0	0.0	0.0	11.0
6	2.0	2.387487625093807	1.7387440276531534	1.0	0.0	0.0	0.0	1.0	0.0	12.0
7	3.0	2.84	2.24	1.0	1.0	1.0	0.0	0.0	0.0	12.0
8	3.0	3.36	2.8	1.0	0.0	0.0	0.0	1.0	1.0	12.0
9	3.0	2.44	1.96	1.0	0.0	1.0	1.0	1.0	0.0	13.0
10	3.0	5.16	4.28	0.0	0.0	0.0	0.0	0.0	0.0	12.0

B.3 Audit Trace: Fold 3 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	1.44	1.0	1.0	0.0	1.0	0.0	1.0	1.0	11.0
2	3.0	6.08	86.3	1.0	0.0	0.0	0.0	0.0	0.0	11.0
3	3.0	2.0457351093544283	1.4166985969886888	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	3.0	1.44	1.0	1.0	0.0	1.0	0.0	1.0	0.0	11.0
5	3.0	6.08	86.3	0.0	0.0	0.0	0.0	0.0	0.0	11.0
6	4.0	2.426623006097093	1.748789728211432	1.0	0.0	0.0	0.0	1.0	0.0	12.0
7	4.0	4.24	3.68	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	3.0	3.87	2.68	0.0	0.0	0.0	0.0	0.0	0.0	12.0
9	5.0	3.67	76.8	0.0	1.0	1.0	0.0	1.0	0.0	12.0
10	3.0	3.8	2.84	1.0	0.0	0.0	0.0	1.0	0.0	12.0

B.4 Audit Trace: Fold 4 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	1.44	1.0	1.0	0.0	1.0	0.0	0.0	1.0	11.0
2	3.0	3.815287073459424	3.072994804851823	1.0	0.0	0.0	1.0	1.0	0.0	11.0
3	3.0	2.1121045074860216	1.4436079753481577	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	3.0	1.44	1.0	1.0	0.0	1.0	0.0	1.0	1.0	12.0
5	3.0	6.3	86.3	0.0	0.0	0.0	0.0	0.0	0.0	11.0
6	4.0	2.6998147371832912	1.9910658346495849	1.0	0.0	0.0	0.0	1.0	0.0	12.0
7	3.0	1.84	1.28	1.0	0.0	0.0	0.0	1.0	1.0	11.0
8	5.0	4.95	4.12	1.0	0.0	0.0	0.0	0.0	1.0	11.0
9	3.0	2.92	2.28	1.0	0.0	0.0	0.0	1.0	0.0	11.0
10	3.0	2.32	1.76	0.0	0.0	0.0	0.0	0.0	0.0	12.0

B.5 Audit Trace: Fold 5 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.539930457648436	2.316798482327397	1.0	0.0	0.0	0.0	1.0	0.0	11.0
2	3.0	3.509890880610297	3.088517774778904 5	1.0	0.0	0.0	1.0	1.0	0.0	11.0
3	3.0	1.895328339161879	1.258574973194690 2	0.0	0.0	0.0	0.0	0.0	0.0	12.0
4	3.0	1.964761445574455 3	1.795989003881186	1.0	0.0	0.0	0.0	1.0	0.0	11.0
5	3.0	1.717345342170025 5	1.719939840397349 6	1.0	0.0	0.0	0.0	1.0	0.0	12.0
6	4.0	5.813615547933812	2.880836827510049 3	0.0	0.0	0.0	0.0	0.0	0.0	11.0
7	3.0	3.0	2.44	1.0	0.0	0.0	0.0	1.0	1.0	12.0
8	3.0	2.4	1.88	1.0	0.0	0.0	0.0	1.0	0.0	11.0
9	3.0	4.36	3.92	1.0	0.0	0.0	0.0	0.0	0.0	11.0
10	3.0	3.4	2.8	1.0	1.0	0.0	0.0	1.0	1.0	11.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.664739751593703 7	2.417372567586074 4	1.0	0.0	0.0	0.0	1.0	0.0	11.0
2	3.0	6.3	86.3	0.0	0.0	0.0	0.0	0.0	0.0	11.0
3	3.0	1.851012352151960 6	1.237393754034128 2	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	3.0	2.075746981970157 3	1.751836330873767	1.0	0.0	0.0	0.0	1.0	0.0	11.0
5	3.0	1.867578749325511	1.717133666043846 3	1.0	0.0	0.0	0.0	1.0	0.0	12.0
6	3.0	6.3	86.3	0.0	0.0	0.0	0.0	1.0	0.0	12.0
7	2.0	3.24	2.52	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	3.0	2.4	1.64	0.0	0.0	0.0	0.0	0.0	0.0	11.0
9	3.0	4.28	4.44	1.0	0.0	0.0	0.0	0.0	0.0	12.0
10	3.0	3.24	52.3	0.0	0.0	0.0	0.0	0.0	0.0	12.0

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	1.44	0.96	1.0	0.0	1.0	0.0	1.0	1.0	12.0
2	3.0	3.92734015899066 82	3.11204993458433 7	1.0	0.0	0.0	1.0	1.0	0.0	11.0
3	3.0	1.81943765529213 03	1.25182817904828 4	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	3.0	1.44	0.96	1.0	0.0	1.0	0.0	1.0	1.0	11.0
5	3.0	1.52640718484121 44	1.68869472577904 48	1.0	0.0	0.0	0.0	1.0	0.0	12.0
6	4.0	4.94112332086912 9	2.98683899203635 5	0.0	0.0	0.0	0.0	0.0	0.0	11.0
7	3.0	2.84	2.16	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	3.0	2.88	2.48	0.0	0.0	0.0	0.0	0.0	0.0	11.0
9	3.0	2.2	1.44	1.0	0.0	0.0	0.0	1.0	0.0	12.0
10	4.0	5.16	4.96	1.0	0.0	0.0	0.0	0.0	0.0	11.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	1.44	0.96	1.0	0.0	0.0	0.0	1.0	0.0	12.0
2	3.0	3.51493264869224 43	2.79352494911129 46	1.0	0.0	0.0	0.0	1.0	0.0	11.0
3	3.0	1.84282902230762 58	1.27803853687365 98	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	3.0	1.44	0.96	1.0	0.0	0.0	0.0	1.0	0.0	12.0
5	3.0	1.47384004753310 26	1.23543298990972 52	1.0	0.0	0.0	0.0	1.0	0.0	12.0
6	4.0	5.71897813751221 1	2.94796250766177 93	0.0	0.0	0.0	0.0	0.0	0.0	11.0
7	3.0	2.76	2.28	0.0	0.0	0.0	0.0	0.0	0.0	11.0
8	4.0	5.03	79.3	1.0	0.0	0.0	1.0	0.0	0.0	11.0
9	2.0	2.16	1.56	1.0	0.0	0.0	0.0	1.0	0.0	11.0
10	3.0	2.0	1.0	1.0	0.0	1.0	0.0	1.0	1.0	11.0

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B.9 Audit Trace: Fold 9 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	6.3	0.96	2.0	0.0	0.0	1.0	1.0	1.0	11.0
2	3.0	3.69913001731708 2	3.04680993925931 84	1.0	0.0	0.0	1.0	1.0	0.0	11.0
3	3.0	1.65294017944756 02	1.09155979534693 98	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	3.0	1.91892305278104 8	1.77111016230821 55	1.0	0.0	0.0	0.0	1.0	0.0	11.0
5	3.0	1.44485613569921 47	1.04825069643113 16	1.0	0.0	0.0	0.0	1.0	0.0	12.0
6	4.0	5.95735695310063 6	3.28051745005588 9	0.0	0.0	0.0	0.0	0.0	0.0	11.0
7	3.0	3.44	3.13	1.0	0.0	0.0	0.0	1.0	1.0	12.0
8	3.0	3.64	3.12	1.0	0.0	0.0	0.0	1.0	0.0	12.0
9	3.0	2.56	60.9	0.0	0.0	0.0	0.0	0.0	0.0	11.0
10	3.0	2.8	3.36	1.0	0.0	0.0	0.0	1.0	0.0	12.0

B.10 Audit Trace: Fold 10 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	1.44	1.0	2.0	0.0	0.0	0.0	1.0	0.0	12.0
2	3.0	3.47030345548259 24	2.71020213830291 9	1.0	0.0	0.0	0.0	1.0	0.0	11.0
3	3.0	2.11901430761217 17	1.49025613525797 66	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	3.0	1.44	1.0	0.0	0.0	1.0	0.0	0.0	0.0	12.0
5	3.0	1.55495917808823 92	1.37321655466795 5	1.0	0.0	0.0	0.0	1.0	0.0	12.0
6	4.0	6.28574660104170 1	3.35673801481576 06	0.0	0.0	0.0	0.0	0.0	0.0	11.0
7	3.0	3.48	2.52	1.0	0.0	0.0	0.0	1.0	0.0	14.0
8	3.0	4.48	3.48	0.0	0.0	0.0	0.0	0.0	0.0	12.0
9	3.0	2.63	67.3	1.0	0.0	0.0	1.0	1.0	0.0	11.0
10	4.0	5.16	4.96	1.0	0.0	0.0	0.0	0.0	0.0	11.0